# Long-Form Midterm (Chapter 4 Focus)

**Submission:**

- Submit a single PDF report (exported from Markdown/Word/LaTeX).
- Submit your **code as a .zip file** (Python/Matlab/etc.) separately through Canvas.
- All plots must be generated computationally (Python/Matplotlib, Matlab, etc.). No hand-drawn plots.
- Annotate axes clearly with units.
  **Structure:**
- Each section must begin on a new page.
- After each plot or computational result, include a **Reflection/Discrepancy Analysis** subsection (see rubric below).
- Final day: **Presentation** (10–15 min per student).

# Task 1 — Intrinsic Carrier Concentration $n_i(T)$

- Derive the expression for $n_i(T)$ for your assigned semiconductor.
- Implement the computation over a temperature range (200 K – 600 K).
- Plot $n_i(T)$ on both linear and logarithmic scales.
- Compare with literature values where possible.

# Task 2 — Fermi Level $E_F(T)$

- Calculate the Fermi level as a function of temperature for your semiconductor.
- Include intrinsic, doped-$n$, and doped-$p$ cases.
- Plot $E_F(T)$ with respect to conduction and valence band edges.
- Discuss limits ($T \to 0$, $T \to \infty$).

# Task 3 — Incomplete Ionization

- Incorporate incomplete ionization of dopants into your model.
- Show the freeze-out, extrinsic, and intrinsic regimes clearly.
- Plot carrier concentration vs. $T$ including incomplete ionization.

# Task 4 — Literature Review and Band Diagram

- Find at least **two references** (journal, textbook, or reliable database).
- Identify one real **device application** for your semiconductor.

- Explain why its properties (band gap, $n_i$, mobility, etc.) make it suitable.
- Include a **band diagram** or **band structure figure** from the literature (with citation).

# Task 5 — Experimental Comparison

- Locate at least one piece of **experimental data** for your semiconductor ($n_i(T)$, $E_g(T)$, mobility, etc.).
- Overlay with your computational result where possible.
- Highlight at least one **agreement** and one **discrepancy**, and explain.

# Reflection and Discrepancy Analysis (15%)

Each section of the midterm must include a **Reflection/Discrepancy** subsection.
This is graded separately from correctness.

## Grading Breakdown

- **Method Explanation (5%)**
  - Briefly describe how you set up your calculation or code (what formula, what loop/solver, what assumptions).
  - Credit is for clarity, not correctness.
- **Expected vs. Actual (5%)**
  - State what you expected the trend/curve to look like (e.g., "$n_i(T)$ should increase exponentially with $T$").
  - Compare this expectation with your computed plot.
  - Highlight any mismatches.
- **Interpretation (5%)**
  - Offer at least one possible reason for discrepancy.
  - This can be physics-based (wrong assumption), math-based (units, constants), or computational (plotting/log scale).

## Reflection Template

After each plot or result, answer:

1. **What did you expect?**
   (Describe the anticipated behavior: shape, limits, trends.)
2. **What did you find?**
   (Summarize the actual computed result or plot.)
3. **If there's a mismatch, why?**
   (Give at least one explanation in 2–3 sentences.)

# Task 6 — General Solution and Superposition

- Summarize how the wavefunction can be written as a superposition of stationary states:

  $\Psi(x,t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}. \$\$ - Discusshowthecoefficients\$|c_n|^2\$aredeterminedbyoccupations$

- For **fermions** (electrons, nuclei): Fermi–Dirac distribution

$$f(E_n) = \frac{1}{e^{(E_n-\mu)/k_BT}+1}.$$

- For **bosons** (phonons, photons): Bose–Einstein distribution

$$n(E) = \frac{1}{e^{(E-\mu)/k_BT}-1}.$$

- In the classical limit ($E-\mu \gg k_BT$), both reduce to Maxwell–Boltzmann:

f(E) \approx e^{-(E-\mu)/k_BT}. $$ ## Task 7 — Final Presentation

- Present your findings to the class (10–15 min).
- Include:
  1. Computational methods (what you coded).
  2. Key plots ($n_i(T)$, $E_F(T)$, incomplete ionization).
  3. Literature review and band diagram.
  4. Comparison with experiment.
  5. Reflection on discrepancies and physical interpretation.

# Long-Form Midterm Report Format

Your report must be submitted as a single PDF.
Each section should start on a **new page**.
Figures must be labeled and referenced in the text.
Keep the write-up concise and technical — avoid filler text.

# 1. Introduction

- Identify your assigned semiconductor.
- State why it is important in semiconductor physics.
- Summarize the **scope of your project** (carrier concentration, Fermi level, ionization, comparison with experiment).
- Include references to textbooks or papers (IEEE style).

# 2. Methods

- Clearly state the **equations used** (with sources).
- Describe **assumptions** (intrinsic vs. extrinsic, complete vs. incomplete ionization).
- Provide short **code snippets** if necessary to illustrate your approach (not the full code).
- Mention the constants and parameters you used ($k_B$, $\hbar$, $E_g$, etc.).

# 3. Results

- Present plots in order:
  1. $n_i(T)$ vs. $T$
  2. $E_F(T)$ for intrinsic and doped cases
  3. Carrier concentration with incomplete ionization
  4. Literature band diagram (with citation)
  5. Comparison with experimental data (overlay if possible)
- Each figure should have a **caption** and be referenced in the text.

# 4. Reflection and Discrepancy Analysis

For each result:

1. What did you expect?
2. What did you find?
3. If there was a mismatch, why?

# 5. Discussion

- Summarize key physics insights (e.g., why $n_i$ rises with $T$, why incomplete ionization matters).
- Compare with literature: how close were your results?
- Comment on limitations of your model (constants, approximations, missing effects).

# 6. Applications

- From your literature review:
  - Describe one device that uses your semiconductor.
  - Explain (briefly) how the material properties make it suitable.

# 7. Conclusion

- Restate your main findings.
- Identify one thing you learned from coding.
- Identify one open question about your material.

# References

- Use IEEE style citations.
- At least **two journal/textbook references** are required.

# Grading Breakdown

- Computational tasks and plots: **45%**
- Literature review and band diagram: **15%**
- Experimental comparison: **10%**
- Reflection and discrepancy analysis: **15%**
- Presentation: **15%**

# Submission Requirements

1. **Final Report (PDF)**
   - Structured according to the Report Format.
   - Figures must be labeled, referenced in the text, and explained.

You must submit a **.zip archive** to Canvas containing:

1. **Code Folder**
   - All scripts used to generate your results.
   - Code must run without modification and reproduce the plots.
   - Organize clearly (e.g., `ni_vs_T.py`, `fermi_level_doping.py`, `incomplete_ionization.py`).
2. **References Folder**
   - PDFs of **every reference you cite** in your report.
   - References from textbooks (can just be the page number)
   - At least one peer-reviewed journal article or reliable source.
   - Files must be named consistently (google scholar name from .bib).
   - the relevant portions you used should be highlighted.

# Midterm Presentation Format

**Time Limit:** 10–15 minutes per student
**Deliverables:** Slides (PDF or PPTX) + Live demonstration of code/plots

## Required Sections

1. **Introduction (1–2 min)**

- Assigned semiconductor + why it's important
- Quick overview of band gap, typical applications

2. **Methods (2–3 min)**
   - Show your code workflow (equations implemented, parameters used)
   - Explain one loop/function/solver in detail
   - Justify any assumptions (e.g., constant effective mass)
3. **Key Results (4–5 min)**
   - $n_i(T)$ (linear + log scale)
   - Fermi level vs. $T$ (intrinsic, n-type, p-type)
   - Incomplete ionization plot
   - At least one comparison with **Si results** (sanity check)
4. **Literature Review & Band Diagram (2–3 min)**
   - Show one band diagram or band structure figure from a cited source
   - Briefly summarize at least one application for your semiconductor
   - Connect properties (band gap, mobility, etc.) to application
5. **Experimental Comparison & Reflection (2–3 min)**
   - Overlay one experimental dataset with your results
   - Identify at least one agreement and one discrepancy
   - Provide a physical/computational explanation
6. **Conclusion (1 min)**
   - Summarize your main finding(s)
   - State one limitation of your model
   - Suggest one next step (e.g., adding temperature dependence of $E_g$)

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## Live Demonstration

- Open your code and re-run one plot (e.g., $n_i(T)$ for Si).
- Be ready to explain *what part of the code controls what part of the physics*.

## Q&A (2–3 min)

- Instructor and classmates may ask probing questions:
  - "Why does $n_i$ increase with $T$?"
  - "Where does the band gap enter your code?"
  - "What changes if you double the doping concentration?"

# Academic Integrity

- A citation in your report without a corresponding PDF in your archive will not be counted.
- AI-generated or fabricated references will result in penalties.
- All references must be **actual documents you read**, not generic Google Scholar links.